

MURALI G

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Professional Summary

Embedded Systems Engineer with hands-on experience in Embedded C, ARM Cortex-M firmware, STM32, ESP32, peripheral driver development, and hardware-software debugging. Experienced with UART, SPI, I2C, CAN, ADC, GPIO, timers, interrupts, FreeRTOS, BLE, and PCB bring-up. Strong foundation in embedded systems, digital electronics, firmware development, and real-time sensor acquisition, with experience developing biomedical and IoT applications.

Work Experience

Embedded Hardware Intern

07/2025 – Present | Chennai

Qneuro India Pvt. Ltd.

- Developed and debugged Embedded C firmware on ARM Cortex-M microcontrollers for real-time EEG acquisition and biomedical data processing.
- Implemented BLE communication for biomedical sensor data transmission, supporting wireless transfer of acquired physiological data.
- Developed an Automatic Gain Control (AGC) algorithm for dynamic IR/Red LED calibration based on skin tone, demonstrating experience with signal acquisition and sensor calibration.
- Integrated Linux-based Raspberry Pi systems for TCP/IP communication and camera streaming and performed PCB bring-up, circuit debugging, and hardware-software integration.
- Conducted firmware testing, calibration, validation, and system verification using oscilloscopes and logic analyzers, collaborating with hardware and software teams throughout product development.

Skills

Programming: Embedded C, Python

Microcontrollers: STM32, ESP32/ESP32-S3, ARM Cortex-M

Communication Protocols: UART, SPI, I2C, BLE, TCP/IP, CAN

RTOS/OS: FreeRTOS, Embedded Linux

Debugging & Testing: Oscilloscope, Logic Analyzer, JTAG/SWD, Hardware-Software Debugging, Firmware Validation

Development Tools: STM32CubeIDE, STM32CubeMX, Git, Altium Designer, VS Code

Projects

Predictive Machine Health and Control Hub

2025

- Developed a real-time sensor acquisition system using STM32 and ESP32 for vibration, temperature, pressure, and current monitoring.
- Integrated cloud connectivity for remote monitoring and predictive maintenance. Developed a web-based dashboard for real-time sensor visualization and equipment monitoring.

Fetal Monitoring System

2024

- Developed an embedded project on ESP32 for motion-based fetal monitoring. Integrated three IMU sensors to improve the accuracy and reliability of fetal movement detection during pregnancy.

Education

Bachelor of Engineering, Biomedical Engineering (Hons.)

2022 – 2026

Bannari Amman Institute of Technology

CGPA - 8.2

Higher Secondary and Secondary Education

2019 – 2022

Nirmala Matriculation Higher Secondary School

HSC - 83.3% | SSLC - 96.4%

Certificates

Radio Frequency Applications – Microchip Technology Inc. | PCB Design – Altium Designer – Altium®

Achievements

Winner – MSME Idea Hackathon 5.0

2026

Government of India

Recognized and awarded ₹6 Lakhs for the project titled “Predictive Machine Health and Control Hub”.

Finalist - BITHack 23

2023

Hardware edition for Home Automation System